

Multi-Dimensional Super Gradient Descent: A Probabilistic Multi-Path Search for Boosted Ensembles

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Abstract

We introduce *Multi-Dimensional Super Gradient Descent* (MDSGD), a training-and-search paradigm that extends gradient-boosted decision trees (GBDT) into a coupled family of ensembles (“universes”) connected across boosting rounds (“time”). A stochastic controller, the *alpha-naut*, traverses this *model graph* with reversible moves—adding/pruning trees (time steps) and jumping across universes—guided by a correlation-based novelty criterion and a tempered acceptance rule. Final predictions are obtained by probability stacking across selected universes, yielding calibrated “multi-versal” probabilities. The framework supports forward forecasting via filtering and backward inference via smoothing (with teacher–student distillation for causal deployment). We present objective terms, jump dynamics, and a practical algorithm, and outline an empirical program for stationary and drifting tabular benchmarks.

1 Introduction

Gradient-boosted decision trees ([1, 2, 6]) dominate tabular prediction, yet classical boosting is greedily stagewise and follows a single path in model space. We propose to maintain *multiple* boosted paths (*universes*) and to allow reversible movement both along the boosting trajectory and across parallel universes. The key ideas are:

1. a **model graph** whose nodes are partially-built ensembles;
2. a **jump policy** that prefers candidates which are both *novel* (low predictive correlation) and *good* (low loss);
3. **stacked probabilities** across universes for calibrated predictions;
4. optional **temporal regularization** and **forward/backward** inference for nonstationary data.

2 Related Work

Our approach connects to (i) diversity in ensembles and stacking [7]; (ii) snapshot/temporal ensembling [5]; (iii) tempered exploration [3]; and (iv) sequential Monte Carlo for state evolution [4]. Unlike bagging or independent model selection, we define *reversible moves over boosted-tree structure* with a novelty-triggered policy and aggregate predictions across universes.

3 Problem Setup and Notation

We consider binary outcomes $y \in \{0, 1\}$ (extension to multiclass/regression is direct) with features $x \in \mathbb{R}^d$. A universe $u \in \{1, \dots, U\}$ at boosting round $m \in \{0, \dots, M\}$ defines predictor

$$f_{u,m}(x) = \sum_{k=1}^m \nu T_{u,k}(x), \quad p_{u,m}(y=1 \mid x) = \sigma(f_{u,m}(x)), \quad (1)$$

where $T_{u,k}$ are CART trees and $\nu > 0$ is the learning rate.

4 Search Dynamics on the Model Graph

State. A node is $s = (u, m)$ with parameters $\theta_{u,m}$ (tree structures and leaf values built so far). Let $\hat{\mathbf{y}}^{(u,m)} = (p_{u,m}(1 \mid x))_{x \in V}$ be predictions over a held-out validation shard V .

Novelty and quality.

$$\text{novelty}((u', m') \mid (u, m)) = 1 - \text{corr}(\hat{\mathbf{y}}^{(u,m)}, \hat{\mathbf{y}}^{(u',m')}), \quad (2)$$

$$\text{quality}(u', m') = \frac{1}{|V|} \sum_{(x,y) \in V} \ell(y, p_{u',m'}(1 \mid x)) + \alpha \frac{m'}{M}, \quad (3)$$

with ℓ the log-loss and $\alpha \geq 0$ a size penalty.

Moves. From (u, m) propose candidates:

1. **Forward:** $(u, m+1)$ (fit one new tree on residuals).
2. **Backward:** $(u, m-1)$ (prune or shrink last tree).
3. **Lateral:** $(u \pm 1, m)$ (copy/align parameters from neighbor universe).
4. **Diagonal:** $(u \pm 1, m \pm 1)$.

Acceptance (tempered, novelty-gated). Let $q_c = \text{quality}(u, m)$ and $q_j = \text{quality}(u', m')$. With temperature $T_u \geq 0$ and novelty threshold $\tau_{\text{nov}} \in [0, 1]$, accept candidate j if

$$\text{novelty}(j \mid c) \geq \tau_{\text{nov}} \quad \text{and} \quad \text{either } q_j \leq q_c - \delta \text{ or with prob } \exp\left(-\frac{q_j - q_c}{T_u}\right), \quad (4)$$

where $\delta \geq 0$ ensures strict improvement when exploiting.

4.1 Alpha-naut Controller (Single or Multiple)

We operate K controllers (alpha-nauts) in parallel with different temperatures and random seeds. Periodically, we attempt *tempered swaps* of states between neighboring temperatures (parallel tempering), improving global exploration.

Algorithm 1: Alpha-naut Multi-Path Search on the Model Graph

Input: Training data $\mathcal{D}$, validation shard V , universes U , max rounds M , temperatures $\{T_u\}$
Initialize U universes with $f_{u,0} \equiv 0$. Set active node $s_u \leftarrow (u, 0)$.
for $t = 1$ **to** T_{outer} **do**
 for each universe u *in parallel* **do**
 $c \leftarrow s_u$; build candidate set $\mathcal{C}(c) = \{(u, m \pm 1), (u \pm 1, m), (u \pm 1, m \pm 1)\}$ (valid indices only).
 for each $j \in \mathcal{C}(c)$ **do**
 Fit/prune as needed to realize f_j ; compute $\text{quality}(j)$ via (3) and $\text{novelty}(j \mid c)$ via (2) (reuse cached predictions when possible).
 end
 Choose the first j satisfying acceptance rule (4); if none, keep c .
 Update $s_u \leftarrow j$ if accepted.
 end
 Every K steps: attempt tempered swaps between neighboring temperatures.
end
Return active set $\{f_{u,m_u}\}$.

5 Multi-versal Probability and Stacking

Given selected universes $\{(u, m_u)\}_{u=1}^U$, we form a calibrated mixture

$$\hat{p}(1 \mid x) = \sum_{u=1}^U w_u p_{u,m_u}(1 \mid x), \quad \mathbf{w} \in \Delta^{U-1}, \quad (5)$$

with weights learned by convex risk minimization on V :

$$\mathbf{w}^* = \arg \min_{\mathbf{w} \in \Delta^{U-1}} \frac{1}{|V|} \sum_{(x,y) \in V} \ell \left(y, \sum_u w_u p_{u,m_u}(1 \mid x) \right). \quad (6)$$

Other calibrators (temperature scaling, isotonic) can be applied post-stacking.

6 Time, Drift, and Bidirectional Inference

Let θ_t denote latent environment state. We consider a state-space view

$$\theta_{t+1} \sim \mathcal{T}(\theta_t), \quad y_t \sim \mathcal{P}(y \mid x_t, \theta_t), \quad (7)$$

with per-universe parameters evolving via online boosting. For **forecasting** (filtering),

$$p(y_{t+h} \mid x_{1:t}) \approx \sum_u w_u \int p_u(y_{t+h} \mid x_{t+h}, \theta_{t+h}) p_u(\theta_{t+h} \mid \mathcal{D}_{1:t}) d\theta_{t+h}. \quad (8)$$

For **retrodiction** (smoothing) using future context,

$$p(\theta_t \mid \mathcal{D}_{1:T}) \propto p(\theta_t \mid \mathcal{D}_{1:t}) \int p(\theta_{t+1} \mid \theta_t) p(\theta_{t+1} \mid \mathcal{D}_{1:T}) d\theta_{t+1}. \quad (9)$$

We implement this with a bidirectional teacher (sees $[t-w, t+w]$) and distill to a causal student for deployment to avoid leakage.

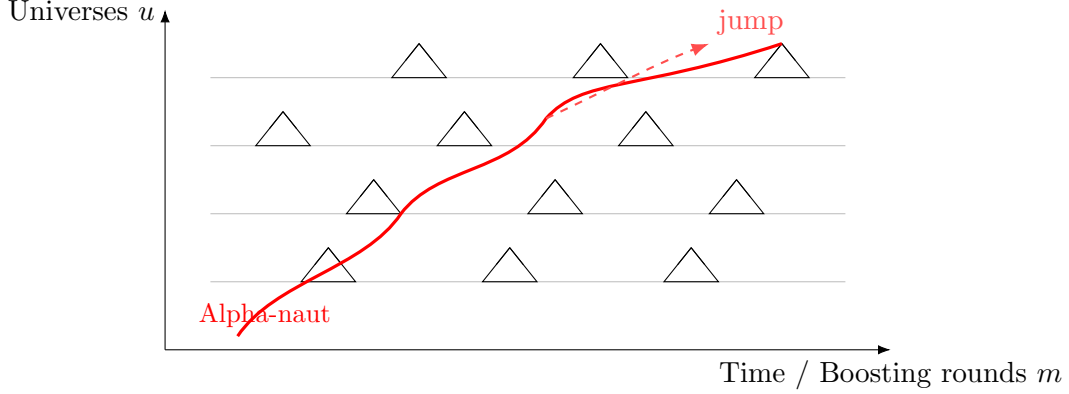


Figure 1: Model graph view: rows are universes u , horizontal axis is boosting round m . The alpha-naut follows a reversible path, occasionally “jumping” to a new universe when the novelty gate passes.

7 A TikZ Schematic

8 Training Details

We build on LightGBM/XGBoost internals for tree fitting. To bound compute:

- share feature histograms across universes at the same m ;
- cache validation predictions to compute correlations cheaply;
- cap the candidate neighborhood to at most six nodes per step: $\{(u, m \pm 1), (u \pm 1, m), (u \pm 1, m \pm 1)\}$.

9 Evaluation Plan

Baselines. Single GBDT, bagged GBDTs, snapshot ensembles, online boosting.

Metrics. AUC/accuracy; log-loss; calibration (Brier, ECE); risk-coverage; post-change-point regret.

Datasets. Stationary (Higgs/Otto); drift (Electricity, Airline, synthetic covariate/label shift).

Ablations. No novelty gate; no backward moves; uniform vs. learned stacking; varying U , temperatures, and swap period.

10 Ethical Considerations

We report calibration and uncertainty with abstention modes; forbid training-time peeking at future labels in deployed models (teacher-student distillation only); and document data sources, drift monitoring, and rollback procedures.

11 Conclusion

MDSGD frames boosted-tree training as a reversible search over a graph of ensembles, using novelty-gated jumps and tempered acceptance, and aggregates universes via calibrated stacking. The approach promises improved exploration, drift robustness, and probabilistic calibration while remaining practical on existing tooling.

References

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